

The Calculus of Structural Resources

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Abstract

We develop the *Calculus of Structural Resources*: a differential, discrete, variational, statistical, and algorithmic calculus for systems that allocate resources, observe them partially, transform them dynamically, consume them over time, and eventually exhaust them. The underlying object is a structural resource system

$$\mathcal{S} = (\Theta, X, R, Y, \Sigma, \alpha, \delta, \omega, \chi),$$

where Θ is a hidden allocation space, X is a state space, R is a resource space, Y is an observation space, $\alpha : \Theta \rightarrow X \times R$ is an allocation map, δ is a resource-consuming transition law, $\omega : X \times R \rightarrow Y$ is an observation map, and $\chi : R \rightarrow \{0, 1\}$ is a resource-availability predicate. The calculus has two central axes. First, ambiguity is measured by fibres and tangent fibres of the observation-allocation map

$$\Omega = \omega \circ \alpha.$$

Second, exhaustion is measured by trajectories reaching a resource boundary

$$h(r) = 0.$$

The core formulas are the structural chain rule

$$D\Omega_t = D\omega_{\phi_t(\alpha(\theta))} D\phi_t|_{\alpha(\theta)} D\alpha_\theta,$$

the tangent ambiguity equation

$$\mathcal{A}_\theta(t) = \ker D\Omega_t(\theta),$$

and the exhaustion-time sensitivity formula

$$D_\theta \tau = -\frac{Dh_{r(\tau)} P_R(\tau)}{Dh_{r(\tau)} F_R(z(\tau), u(\tau))}.$$

We prove the main differential and discrete theorems, develop algorithms for tangent ambiguity, lifetime sensitivity, silent consumption, crisis detection, and resource-aware learning, and provide applications to economics, finance, political science, geopolitics, international relations, warfare economics, warfare, statistics, data science, and artificial intelligence.

The treatise ends with “The End”

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1 Introduction

Many systems are neither fully visible nor indefinitely sustainable. They begin with hidden allocations, produce partial observations, evolve under rules, consume resources, and may eventually reach a terminal boundary. Examples include:

a computation consuming memory,
a firm consuming liquidity,
an army consuming ammunition and logistics,
a state consuming legitimacy and fiscal capacity,
a learning algorithm consuming samples and compute,
a thermodynamic machine consuming free energy.

The theory of structural resources studies the object:

$$\mathcal{S} = (\Theta, X, R, Y, \Sigma, \alpha, \delta, \omega, \chi).$$

The calculus developed here studies how such systems change locally and globally.

The guiding idea is:

structure allocates, observation forgets, dynamics consumes, exhaustion terminates.

This treatise develops a formal calculus with the following primitives:

$$D\alpha, \quad D\omega, \quad D\Omega, \quad \ker D\Omega, \quad \dot{r}, \quad c = -\dot{r}, \quad h(r) = 0, \quad D_\theta \tau.$$

The main slogan is:

Structural Resource Calculus differentiates ambiguity and integrates exhaustion.

2 Structural Resource Systems

Definition 2.1 (Structural resource system). A structural resource system is a tuple

$$\mathcal{S} = (\Theta, X, R, Y, \Sigma, \alpha, \delta, \omega, \chi),$$

where:

1. Θ is the hidden allocation space;
2. X is the state space;
3. R is the resource space;
4. Y is the observation space;
5. Σ is the action or input space;
6. $\alpha : \Theta \rightarrow X \times R$ is the allocation map;
7. $\delta : X \times R \times \Sigma \rightarrow X \times R$ is a transition map;
8. $\omega : X \times R \rightarrow Y$ is an observation map;
9. $\chi : R \rightarrow \{0, 1\}$ is the availability predicate.

The structural state is

$$z = (x, r) \in X \times R.$$

The allocation map is written

$$\alpha(\theta) = (\alpha_X(\theta), \alpha_R(\theta)).$$

The observation-allocation map is

$$\Omega = \omega \circ \alpha : \Theta \rightarrow Y.$$

Definition 2.2 (Available and exhausted regions). Define

$$R^{\text{av}} = \chi^{-1}(1), \quad R^{\text{un}} = \chi^{-1}(0).$$

The evolving and exhausted regions of $X \times R$ are

$$Q^{\text{ev}} = X \times R^{\text{av}}, \quad Q^{\text{ex}} = X \times R^{\text{un}}.$$

Definition 2.3 (Smooth structural resource system). A smooth structural resource system is a structural resource system in which Θ, X, R, Y are smooth manifolds, α and ω are smooth maps, and the dynamics are given by a vector field

$$\dot{z}(t) = F(z(t), u(t)), \quad z(t) = (x(t), r(t)).$$

Writing

$$F = (F_X, F_R),$$

we have

$$\begin{aligned} \dot{x}(t) &= F_X(x(t), r(t), u(t)), \\ \dot{r}(t) &= F_R(x(t), r(t), u(t)). \end{aligned}$$

3 Allocation, Observation, and Ambiguity

3.1 Fibre Ambiguity

Definition 3.1 (Allocation fibre). For an observation $y \in Y$, the allocation fibre is

$$F_y = \Omega^{-1}(y) = \{\theta \in \Theta : \omega(\alpha(\theta)) = y\}.$$

Definition 3.2 (Structural ambiguity). The observation y is structurally ambiguous if

$$|F_y| > 1.$$

It is structurally determined if

$$|F_y| = 1.$$

It is impossible under the model if

$$|F_y| = 0.$$

Definition 3.3 (Ambi-object induced by a fibre). If

$$F_y = \{\theta_1, \dots, \theta_n\},$$

then the associated ambi-object is

$$A_y = \langle \theta_1 \mid \dots \mid \theta_n \rangle.$$

Theorem 3.4 (Fibre ambiguity theorem). *Ambiguity is exactly non-injectivity of the observation-allocation map Ω over the observed value. In particular, a structural resource system is globally identifiable if and only if*

$$\Omega : \Theta \rightarrow Y$$

is injective.

Proof. For $y \in \text{im}(\Omega)$, the set of hidden allocations compatible with y is exactly

$$F_y = \Omega^{-1}(y).$$

If Ω is injective, every fibre over its image has exactly one element, so every observable value identifies a unique hidden allocation. Conversely, if every fibre over $\text{im}(\Omega)$ has exactly one element, then whenever

$$\Omega(\theta_1) = \Omega(\theta_2),$$

both θ_1 and θ_2 lie in the same singleton fibre, so $\theta_1 = \theta_2$. Thus Ω is injective. $\square$

3.2 Tangent Ambiguity

Definition 3.5 (Allocation differential). For a smooth allocation map

$$\alpha : \Theta \rightarrow X \times R,$$

the allocation differential at θ is

$$D\alpha_\theta : T_\theta\Theta \rightarrow T_{\alpha(\theta)}(X \times R).$$

Since

$$\alpha(\theta) = (\alpha_X(\theta), \alpha_R(\theta)),$$

we write

$$D\alpha_\theta = \begin{pmatrix} D\alpha_X(\theta) \\ D\alpha_R(\theta) \end{pmatrix}.$$

Definition 3.6 (Observation differential). The observation differential is

$$D\omega_z : T_z(X \times R) \rightarrow T_{\omega(z)}Y.$$

For $z = (x, r)$, it decomposes as

$$D\omega_z(\delta x, \delta r) = D_x\omega_z \delta x + D_r\omega_z \delta r.$$

Theorem 3.7 (Structural chain rule). *Let*

$$\Omega = \omega \circ \alpha.$$

Then

$$D\Omega_\theta = D\omega_{\alpha(\theta)} \circ D\alpha_\theta.$$

Proof. This is the ordinary chain rule for smooth maps applied to the composition

$$\Theta \xrightarrow{\alpha} X \times R \xrightarrow{\omega} Y.$$

□

Definition 3.8 (Tangent ambiguity space). The tangent ambiguity space at θ is

$$\mathcal{A}_\theta = \ker D\Omega_\theta.$$

Thus

$$\mathcal{A}_\theta = \{v \in T_\theta\Theta : D\Omega_\theta v = 0\}.$$

Theorem 3.9 (Rank formula for local ambiguity). *If $D\Omega_\theta$ has finite rank, then*

$$\dim \mathcal{A}_\theta = \dim \Theta - \text{rank } D\Omega_\theta.$$

Proof. The linear map

$$D\Omega_\theta : T_\theta\Theta \rightarrow T_{\Omega(\theta)}Y$$

has kernel $\mathcal{A}_\theta$. By rank-nullity,

$$\dim T_\theta\Theta = \text{rank } D\Omega_\theta + \dim \ker D\Omega_\theta.$$

Since $\dim T_\theta\Theta = \dim \Theta$, the result follows. □

Corollary 3.10 (Local identifiability). *The allocation θ is locally identifiable only if*

$$\ker D\Omega_\theta = \{0\}.$$

Equivalently,

$$\text{rank } D\Omega_\theta = \dim \Theta.$$

4 Resource Dynamics and Consumption

Definition 4.1 (Resource velocity). For a smooth structural resource system

$$\dot{z} = F(z, u), \quad z = (x, r),$$

the resource velocity is

$$\dot{r} = F_R(x, r, u).$$

Definition 4.2 (Consumption). If $R = \mathbb{R}_{\geq 0}^k$, the consumption vector is

$$c(t) = -\dot{r}(t).$$

Productive consumption satisfies

$$c(t) \in \mathbb{R}_{\geq 0}^k.$$

Definition 4.3 (Total consumption). The total consumption over $[0, T]$ is

$$C[0, T] = \int_0^T c(t) dt.$$

For scalar resource,

$$C[0, T] = r(0) - r(T).$$

Definition 4.4 (Availability function). A smooth availability function is a map

$$h : R \rightarrow \mathbb{R}$$

such that

$$R^{\text{av}} = \{r : h(r) > 0\}, \quad R^{\text{ex}} = \{r : h(r) = 0\}.$$

The exhaustion boundary is

$$\partial R_{\text{ex}} = \{r : h(r) = 0\}.$$

Definition 4.5 (Exhaustion time). For a trajectory $z(t) = (x(t), r(t))$, the exhaustion time is

$$\tau(z_0, u) = \inf\{t \geq 0 : h(r(t)) = 0\}.$$

If no such time exists, set $\tau = \infty$.

Proposition 4.6 (Scalar exhaustion law). *If*

$$\dot{r}(t) = -c(t), \quad c(t) > 0,$$

and $h(r) = r$, then exhaustion time τ satisfies

$$r(0) = \int_0^\tau c(t) dt.$$

If $c(t) = c > 0$ is constant, then

$$\tau = \frac{r(0)}{c}.$$

Proof. Since $\dot{r} = -c$,

$$r(\tau) - r(0) = - \int_0^\tau c(t) dt.$$

At exhaustion $r(\tau) = 0$. Hence

$$r(0) = \int_0^\tau c(t) dt.$$

If $c(t) = c$, then

$$r(0) = c\tau,$$

so

$$\tau = r(0)/c.$$

□

5 Dynamic Structural Resource Calculus

5.1 Flow and Observation

Let ϕ_t be the flow generated by

$$\dot{z} = F(z, u).$$

The time- t observation-allocation map is

$$\Omega_t = \omega \circ \phi_t \circ \alpha.$$

Theorem 5.1 (Dynamic structural chain rule). *For a smooth flow ϕ_t ,*

$$D\Omega_t(\theta) = D\omega_{\phi_t(\alpha(\theta))} \circ D\phi_t|_{\alpha(\theta)} \circ D\alpha_\theta.$$

Proof. The map Ω_t is a composition:

$$\Theta \xrightarrow{\alpha} X \times R \xrightarrow{\phi_t} X \times R \xrightarrow{\omega} Y.$$

Applying the chain rule to this triple composition gives the stated formula. $\square$

Definition 5.2 (Dynamic tangent ambiguity). The tangent ambiguity space at time t is

$$\mathcal{A}_\theta(t) = \ker D\Omega_t(\theta).$$

The local ambiguity dimension is

$$a(t, \theta) = \dim \ker D\Omega_t(\theta).$$

Corollary 5.3 (Dynamic ambiguity rank formula). *If $D\Omega_t(\theta)$ has finite rank, then*

$$a(t, \theta) = \dim \Theta - \text{rank } D\Omega_t(\theta).$$

Proof. Apply rank-nullity to the linear map $D\Omega_t(\theta)$. $\square$

5.2 Observation Velocity

Proposition 5.4 (Observation velocity). *If*

$$y(t) = \omega(z(t)), \quad \dot{z}(t) = F(z(t), u(t)),$$

then

$$\dot{y}(t) = D\omega_{z(t)} F(z(t), u(t)).$$

Proof. Differentiate $y(t) = \omega(z(t))$ with respect to t . The result follows from the chain rule. $\square$

Definition 5.5 (Silent consumption). A trajectory exhibits silent consumption on an interval I if

$$\dot{y}(t) = 0$$

but

$$\dot{r}(t) \neq 0$$

for $t \in I$. Equivalently,

$$D\omega_z F(z, u) = 0, \quad F_R(z, u) \neq 0.$$

Proposition 5.6 (Silent consumption criterion). *A smooth trajectory has silent consumption at time t if and only if*

$$F(z(t), u(t)) \in \ker D\omega_{z(t)}$$

and

$$F_R(z(t), u(t)) \neq 0.$$

Proof. The equality $F(z(t), u(t)) \in \ker D\omega_{z(t)}$ is equivalent to

$$D\omega_{z(t)}F(z(t), u(t)) = 0,$$

which is equivalent to $\dot{y}(t) = 0$. The second condition is exactly $\dot{r}(t) \neq 0$. Together these are the definition of silent consumption. $\square$

6 Exhaustion-Time Sensitivity

6.1 Variational Equation

Let

$$z(t; \theta)$$

solve

$$\dot{z} = F(z, u), \quad z(0) = \alpha(\theta).$$

Define the sensitivity matrix

$$P(t) = \frac{\partial z(t; \theta)}{\partial \theta}.$$

Theorem 6.1 (Variational equation). *The sensitivity matrix satisfies*

$$\dot{P}(t) = D_z F(z(t), u(t))P(t), \quad P(0) = D\alpha_\theta.$$

Proof. Differentiate the differential equation

$$\dot{z}(t; \theta) = F(z(t; \theta), u(t))$$

with respect to θ . Assuming u is fixed with respect to θ ,

$$\frac{d}{dt} \frac{\partial z(t; \theta)}{\partial \theta} = D_z F(z(t), u(t)) \frac{\partial z(t; \theta)}{\partial \theta}.$$

The initial condition is

$$P(0) = \frac{\partial z(0; \theta)}{\partial \theta} = D\alpha_\theta.$$

$\square$

Writing

$$P(t) = \begin{pmatrix} P_X(t) \\ P_R(t) \end{pmatrix},$$

the matrix $P_R(t)$ is the resource sensitivity.

6.2 Sensitivity Formula

Theorem 6.2 (Exhaustion-time sensitivity). *Suppose exhaustion occurs at finite time $\tau(\theta)$, so*

$$h(r(\tau(\theta), \theta)) = 0.$$

Assume transversality:

$$Dh_{r(\tau)}F_R(z(\tau), u(\tau)) \neq 0.$$

Then τ is differentiable with respect to θ and

$$D_\theta \tau = -\frac{Dh_{r(\tau)}P_R(\tau)}{Dh_{r(\tau)}F_R(z(\tau), u(\tau))}.$$

Proof. The exhaustion boundary condition is

$$h(r(\tau(\theta), \theta)) = 0.$$

Differentiate with respect to θ :

$$Dh_{r(\tau)} \left[\frac{\partial r(\tau, \theta)}{\partial \theta} + \dot{r}(\tau, \theta) D_\theta \tau \right] = 0.$$

Since

$$\frac{\partial r(\tau, \theta)}{\partial \theta} = P_R(\tau), \quad \dot{r}(\tau, \theta) = F_R(z(\tau), u(\tau)),$$

we obtain

$$Dh_{r(\tau)}P_R(\tau) + Dh_{r(\tau)}F_R(z(\tau), u(\tau))D_\theta \tau = 0.$$

Solving for $D_\theta \tau$ gives

$$D_\theta \tau = -\frac{Dh_{r(\tau)}P_R(\tau)}{Dh_{r(\tau)}F_R(z(\tau), u(\tau))}.$$

The transversality condition ensures that the denominator is nonzero. □

Remark 6.3. If

$$Dh_{r(\tau)}F_R(z(\tau), u(\tau)) = 0,$$

then the trajectory touches the exhaustion boundary tangentially. In such cases exhaustion time may be non-differentiable or unstable under perturbation.

7 Discrete Calculus of Structural Resources

7.1 Discrete Resource Differences

In a finite-state or discrete-time system,

$$z_t = (x_t, r_t), \quad z_{t+1} = \delta(z_t, a_t).$$

Define

$$\Delta r_t = r_{t+1} - r_t.$$

Consumption is

$$c_t = r_t - r_{t+1} = -\Delta r_t.$$

Total consumption over T steps is

$$C_T = \sum_{t=0}^{T-1} c_t.$$

Proposition 7.1 (Discrete resource balance). *If $R = \mathbb{N}$ or $R = \mathbb{R}_{\geq 0}$, then*

$$r_T = r_0 - \sum_{t=0}^{T-1} c_t.$$

Proof. Since

$$c_t = r_t - r_{t+1},$$

summing over $t = 0, \dots, T-1$ gives

$$\sum_{t=0}^{T-1} c_t = \sum_{t=0}^{T-1} (r_t - r_{t+1}) = r_0 - r_T.$$

Rearranging yields the claim. □

7.2 Discrete Structural Derivative

Definition 7.2 (Discrete structural derivative). For a scalar function

$$V : X \times R \rightarrow \mathbb{R}$$

and action a , define

$$D_a V(z) = V(\delta(z, a)) - V(z).$$

Definition 7.3 (Discrete resource monotone). A function $V : X \times R \rightarrow \mathbb{R}$ is a discrete resource monotone if

$$D_a V(z) \leq 0$$

for every admissible action a . It is strict if

$$D_a V(z) < 0$$

for every productive transition.

Theorem 7.4 (Discrete exhaustion by strict monotone). *If a finite discrete structural resource system admits a strict resource monotone V with values in a finite well-ordered set, then every productive trajectory is finite.*

Proof. An infinite productive trajectory would produce a strictly decreasing sequence

$$V(z_0) > V(z_1) > V(z_2) > \dots$$

inside a finite well-ordered set. Such a sequence cannot exist. Hence every productive trajectory is finite. □

7.3 Lifetime Bellman Equations

Definition 7.5 (Maximum remaining life). For a discrete system, the maximum remaining life is

$$L(z) = \begin{cases} 0, & z \in Q^{\text{ex}}, \\ 1 + \max_{a: \delta(z, a) \text{ defined}} L(\delta(z, a)), & z \in Q^{\text{ev}}. \end{cases}$$

Definition 7.6 (Minimum forced life). The minimum forced life is

$$M(z) = \begin{cases} 0, & z \in Q^{\text{ex}}, \\ 1 + \min_{a: \delta(z, a) \text{ defined}} M(\delta(z, a)), & z \in Q^{\text{ev}}. \end{cases}$$

Theorem 7.7 (Bellman lifetime theorem). *If the reachable evolving graph is acyclic, then L and M are well-defined and satisfy the above recursions uniquely.*

Proof. Since the reachable evolving graph is acyclic, its nodes admit a reverse topological ordering. For exhausted states, $L = M = 0$. Processing evolving states in reverse topological order, all successor values have already been computed. Therefore the recursions determine L and M uniquely. $\square$

8 Min-Max Calculus of Ambiguous Lifetimes

Definition 8.1 (Ambiguous lifetime profile). Let

$$A = \langle z_1 \mid \cdots \mid z_n \rangle$$

be an ambiguous structural state. Its lifetime profile is

$$\Lambda(A) = \langle \tau_1 \mid \cdots \mid \tau_n \rangle,$$

where

$$\tau_i = \tau(z_i).$$

Definition 8.2 (Existential and universal lifetimes). Define

$$\tau_{\exists}(A) = \min_i \tau_i, \quad \tau_{\forall}(A) = \max_i \tau_i.$$

Theorem 8.3 (Min-max lifetime theorem). *For an ambiguous state A , $\tau_{\exists}(A)$ is the first time at which at least one branch exhausts, and $\tau_{\forall}(A)$ is the first time by which all branches have exhausted.*

Proof. Branch i exhausts at time τ_i . The first branch exhaustion occurs at the minimum of the τ_i . All branches have exhausted exactly when the longest-lived branch has exhausted, namely at the maximum of the τ_i . $\square$

Proposition 8.4 (Differential of unique min-max branches). *If the minimizer i^* of $\tau_{\exists}$ is unique, then*

$$D\tau_{\exists} = D\tau_{i^*}.$$

If the maximizer j^ of $\tau_{\forall}$ is unique, then*

$$D\tau_{\forall} = D\tau_{j^*}.$$

Proof. In a neighbourhood where the minimizer remains unique,

$$\tau_{\exists} = \tau_{i^*}.$$

Therefore their differentials agree. The same argument applies to the unique maximizer. $\square$

Remark 8.5. At ties, min and max are nonsmooth. The appropriate calculus is subdifferential calculus:

$$\partial\tau_{\exists} = \text{conv}\{D\tau_i : \tau_i = \tau_{\exists}\},$$

$$\partial\tau_{\forall} = \text{conv}\{D\tau_i : \tau_i = \tau_{\forall}\}.$$

9 Resource Control and Variational Calculus

9.1 Structural Resource Control

Let

$$\dot{x} = f(x, r, u), \quad \dot{r} = -c(x, r, u),$$

where $u(t) \in U$ is a control.

A standard resource-control problem is:

$$\max_{u(\cdot)} \int_0^T U(x(t), r(t), u(t)) dt$$

subject to

$$r(t) \geq 0, \quad r(0) = r_0.$$

Definition 9.1 (Structural Hamiltonian). The Hamiltonian is

$$H(x, r, u, p_x, p_r) = U(x, r, u) + p_x f(x, r, u) - p_r c(x, r, u).$$

Proposition 9.2 (First-order optimality conditions). *For an interior optimal control, the necessary conditions are*

$$\begin{aligned} u^*(t) &\in \arg \max_u H(x, r, u, p_x, p_r), \\ \dot{p}_x &= -\frac{\partial H}{\partial x}, \quad \dot{p}_r = -\frac{\partial H}{\partial r}. \end{aligned}$$

Proof. These are the Pontryagin necessary conditions applied to the state equations

$$\dot{x} = f(x, r, u), \quad \dot{r} = -c(x, r, u).$$

The signs follow from including the resource equation in the Hamiltonian as

$$p_r \dot{r} = -p_r c(x, r, u).$$

□

Remark 9.3. The costate p_r is the shadow value of one additional unit of resource. In economics and finance, it is the marginal value of liquidity, collateral, fiscal space, or trust. In warfare, it is the marginal value of ammunition, logistics, manpower, or time.

9.2 Hamilton-Jacobi-Bellman Equation

Let $V(x, r)$ be the maximum achievable value before exhaustion.

Theorem 9.4 (Structural resource HJB equation). *If V is smooth and the system is continuous-time Markovian, then V satisfies*

$$0 = \max_u [U(x, r, u) + \nabla_x V \cdot f(x, r, u) - \nabla_r V \cdot c(x, r, u)]$$

on the available region, with boundary condition

$$V(x, r) = 0 \quad \text{when } h(r) = 0.$$

For pure lifetime maximization, $U = 1$.

Proof. The dynamic programming principle over a short interval Δt gives

$$V(x, r) = \max_u [U(x, r, u)\Delta t + V(x + \dot{x}\Delta t, r + \dot{r}\Delta t)].$$

Expanding V to first order, substituting $\dot{x} = f$ and $\dot{r} = -c$, cancelling $V(x, r)$, dividing by Δt , and taking $\Delta t \rightarrow 0$ gives the equation. The boundary condition states that no continuation value remains at exhaustion. □

10 Statistical Structural Resource Calculus

10.1 Random Allocations

Let θ be random with distribution P_Θ . The observed variable is

$$Y = \Omega(\theta).$$

Definition 10.1 (Posterior allocation fibre). Given observation y , the posterior allocation law is

$$P(\theta \in A \mid Y = y).$$

Its support is contained in

$$F_y = \Omega^{-1}(y).$$

Definition 10.2 (Fibre entropy). For a finite fibre F_y , define the fibre entropy

$$H(F_y) = - \sum_{\theta_i \in F_y} p_i(y) \log p_i(y),$$

where

$$p_i(y) = P(\theta = \theta_i \mid Y = y).$$

Remark 10.3. Structural ambiguity is the existence of multiple compatible branches. Fibre entropy measures how probability mass is distributed across those branches. Thus ambiguity is structural; entropy is statistical.

10.2 Fisher Information and Tangent Ambiguity

Suppose Y has likelihood $p(y \mid \theta)$. The Fisher information matrix is

$$I(\theta) = \mathbb{E}_\theta \left[\nabla_\theta \log p(Y \mid \theta) \nabla_\theta \log p(Y \mid \theta)^\top \right].$$

Definition 10.4 (Statistical tangent ambiguity). The statistical tangent ambiguity space is

$$\mathcal{A}_\theta^{\text{stat}} = \ker I(\theta).$$

Proposition 10.5 (Information-null ambiguity). *A direction $v \in T_\theta \Theta$ lies in $\ker I(\theta)$ if and only if infinitesimal motion in direction v is statistically invisible to first order in the score norm.*

Proof. Since $I(\theta)$ is positive semidefinite,

$$v^\top I(\theta) v = \mathbb{E}_\theta \left[(v^\top \nabla_\theta \log p(Y \mid \theta))^2 \right].$$

This quantity is zero if and only if

$$v^\top \nabla_\theta \log p(Y \mid \theta) = 0$$

almost surely. Hence the score has no first-order sensitivity in direction v . □

11 Data Science and AI/ML

11.1 Learning as Ambiguity Reduction Under Resource Consumption

In supervised learning, let:

$$\Theta = H$$

be the hypothesis class,

$$X$$

be the model state,

$$R$$

be sample, compute, memory, or token budget, and

$$Y$$

be the observed training or validation behaviour.

The map

$$\Omega : H \rightarrow Y$$

sends a hypothesis to its observed behaviour. The fibre

$$F_y = \{h \in H : \Omega(h) = y\}$$

is the set of hypotheses observationally indistinguishable at y .

Learning reduces ambiguity if

$$\dim \ker D\Omega_t$$

decreases over time.

Definition 11.1 (Learning efficiency). Define ambiguity dimension

$$a(t) = \dim \ker D\Omega_t.$$

If $c(t) > 0$ is resource consumption, define

$$\eta_{\text{learn}}(t) = \frac{-\dot{a}(t)}{c(t)}.$$

If

$$\eta_{\text{learn}}(t) > 0,$$

the learner reduces ambiguity per unit resource. If

$$\eta_{\text{learn}}(t) = 0,$$

resources are consumed without reducing ambiguity. If

$$\eta_{\text{learn}}(t) < 0,$$

the system consumes resources while becoming less identifiable.

11.2 Version-Space Exhaustion

Let $V_t \subseteq H$ be the version space after data batches $D_1, \dots, D_t$:

$$V_t = \{h \in V_{t-1} : \ell(h, D_t) \leq \epsilon\}.$$

Proposition 11.2 (Version-space monotonicity). *The version spaces satisfy*

$$V_t \subseteq V_{t-1}.$$

Proof. By definition, V_t is obtained from V_{t-1} by imposing an additional constraint. Therefore every element of V_t belongs to V_{t-1} . $\square$

If $V_t = \emptyset$, the hypothesis resource is exhausted inconsistently. If

$$|V_t| = 1,$$

ambiguity is resolved. If the resource budget reaches zero while

$$|V_t| > 1,$$

the system ends in exhausted ambiguity.

12 Economics and Finance

12.1 Balance-Sheet Ambiguity

In finance, let:

Θ = hidden balance-sheet allocations,

X = institutional state,

R = liquidity, collateral, credit, trust,

Y = prices, spreads, ratings, reports.

The map

$$\Omega = \omega \circ \alpha$$

sends hidden balance-sheet structures to observed market signals.

Definition 12.1 (Financial ambiguity space). At allocation θ , the financial ambiguity space is

$$\mathcal{A}_\theta^{\text{fin}} = \ker D\Omega_\theta.$$

It consists of hidden balance-sheet perturbations invisible to market observations.

12.2 Liquidity Exhaustion

Let $r(t)$ be liquidity and let

$$\dot{r}(t) = -c(t).$$

If $c(t) > 0$, liquidity exhaustion occurs when

$$r(\tau) = 0.$$

The exhaustion time is

$$\tau = \inf\{t : r(t) = 0\}.$$

Definition 12.2 (Financial silent consumption). A financial system has silent consumption if

$$\dot{y}(t) = 0$$

for prices or reported indicators, but

$$\dot{r}(t) < 0$$

for liquidity, collateral, or trust.

Definition 12.3 (Structural crisis index). Let

$$a(t) = \dim \ker D\Omega_t$$

and let $h(r(t)) > 0$ be available resource. Define

$$\mathcal{C}(t) = a(t) \frac{-\dot{h}(r(t))}{h(r(t)) + \varepsilon},$$

where $\varepsilon > 0$.

A crisis is structurally severe when ambiguity is high and depletion is fast:

$$a(t) \gg 0, \quad -\dot{h}(r(t)) \gg 0, \quad h(r(t)) \approx 0.$$

13 Political Science, Geopolitics, and International Relations

13.1 Political Capacity as Resource

In political systems, resources include:

legitimacy, fiscal capacity, bureaucratic capacity, coalitional support, attention, coercive capacity.

Let

$$r(t) \in R$$

represent political capacity. Exhaustion occurs when the state cannot continue its chosen policy path:

$$h(r(t)) = 0.$$

13.2 Strategic Ambiguity

In international relations, hidden allocations include:

Θ = force posture, resolve, alliance commitments, industrial capacity, intelligence.

Observations include:

Y = signals, troop movements, speeches, market reactions, diplomatic cables.

The ambiguity fibre

$$F_y = \Omega^{-1}(y)$$

is the set of strategic postures compatible with observed signals.

Definition 13.1 (Strategic ambiguity dimension). The strategic ambiguity dimension is

$$a_{\text{IR}}(t) = \dim \ker D\Omega_t.$$

High $a_{\text{IR}}(t)$ means observed signals do not identify the underlying strategic allocation.

13.3 Escalation as Resource Curvature

Let $r(t)$ be a vector of diplomatic, fiscal, industrial, and military capacity. Resource curvature is

$$\ddot{r}(t).$$

If consumption is

$$c(t) = -\dot{r}(t),$$

then accelerating consumption is

$$\dot{c}(t) > 0.$$

A geopolitical crisis may be characterized by:

$$a_{\text{IR}}(t) \text{ high,} \quad h(r(t)) \text{ low,} \quad \dot{c}(t) > 0.$$

14 Warfare, Warfare Economics, and Logistics

14.1 Warfare as Structural Resource Dynamics

In warfare, take:

X = battlefield state,

R = manpower, ammunition, fuel, logistics, morale, intelligence, time,

Y = observed battlefield reports,

Θ = hidden force allocation and intent.

The dynamics are

$$\dot{x} = F_X(x, r, u, v),$$

$$\dot{r} = F_R(x, r, u, v),$$

where u is friendly action and v is adversary action.

Definition 14.1 (Combat resource burn). The burn rate is

$$c(t) = -F_R(x(t), r(t), u(t), v(t)).$$

14.2 Culminating Point

A force reaches culmination when it can no longer continue offensive operations. In this calculus, culmination is an exhaustion boundary:

$$h(r(t)) = 0.$$

Definition 14.2 (Culmination time). The culmination time is

$$\tau_{\text{culm}} = \inf\{t : h(r(t)) = 0\}.$$

Proposition 14.3 (Logistical sensitivity of culmination). *If the exhaustion boundary is transverse, then the sensitivity of culmination time to initial logistics θ is*

$$D_{\theta}\tau_{\text{culm}} = -\frac{Dh_{r(\tau)}P_R(\tau)}{Dh_{r(\tau)}F_R(z(\tau), u(\tau), v(\tau))}.$$

Proof. This is the exhaustion-time sensitivity theorem applied to the warfare resource vector. $\square$

15 Algorithms

Algorithm 15.1 (Tangent ambiguity computation). Input: smooth maps $\alpha : \Theta \rightarrow X \times R$, $\omega : X \times R \rightarrow Y$, and point θ .

Step	Operation
1	Compute $D\alpha_\theta$.
2	Compute $D\omega_{\alpha(\theta)}$.
3	Form $D\Omega_\theta = D\omega_{\alpha(\theta)}D\alpha_\theta$.
4	Compute a basis for $\ker D\Omega_\theta$.
5	Return $\mathcal{A}_\theta = \ker D\Omega_\theta$ and $a(\theta) = \dim \mathcal{A}_\theta$.

Correctness. By the structural chain rule,

$$D\Omega_\theta = D\omega_{\alpha(\theta)}D\alpha_\theta.$$

The tangent ambiguity space is defined as $\ker D\Omega_\theta$. Therefore computing the kernel of the matrix formed in Step 3 returns exactly the tangent ambiguity space. $\square$

Algorithm 15.2 (Exhaustion-time sensitivity). Input: vector field F , allocation $\alpha(\theta)$, resource boundary $h(r) = 0$, control $u(t)$.

Step	Operation
1	Simulate $\dot{z} = F(z, u)$ from $z(0) = \alpha(\theta)$.
2	Find τ such that $h(r(\tau)) = 0$.
3	Solve $\dot{P} = D_z F(z(t), u(t))P$, $P(0) = D\alpha_\theta$.
4	Extract $P_R(\tau)$.
5	Compute $D_\theta \tau = -\frac{Dh_{r(\tau)}P_R(\tau)}{Dh_{r(\tau)}F_R(z(\tau), u(\tau))}$.
6	Return $D_\theta \tau$.

Correctness. Steps 1 and 2 compute the exhaustion event. Step 3 computes the variational sensitivity matrix by the variational equation. Step 5 applies the exhaustion-time sensitivity theorem. Thus the returned value is the derivative of exhaustion time with respect to allocation. $\square$

Algorithm 15.3 (Silent consumption detection). Input: trajectory $z(t)$, vector field F , observation map ω , thresholds $\varepsilon_y, \varepsilon_r > 0$.

Step	Operation
1	Compute $\dot{y}(t) = D\omega_{z(t)}F(z(t), u(t))$.
2	Compute $\dot{r}(t) = F_R(z(t), u(t))$.
3	If $\ \dot{y}(t)\ \leq \varepsilon_y$ and $\ \dot{r}(t)\ > \varepsilon_r$, flag t .
4	Merge adjacent flagged times into intervals.
5	Return silent-consumption intervals.

Correctness. The criterion for silent consumption is $\dot{y} = 0$ and $\dot{r} \neq 0$. With numerical thresholds, Steps 1–3 test exactly the approximate version of this criterion. Step 4 merely groups detected times into intervals. $\square$

Algorithm 15.4 (Resource-aware learning). Input: hypothesis class H , data batches D_t , loss threshold ϵ , resource budget r_0 .

Step	Operation
1	Initialize $V_0 = H$, $r = r_0$.
2	For $t = 1, 2, \dots$ while $r > 0$:
3	$V_t = \{h \in V_{t-1} : \ell(h, D_t) \leq \epsilon\}$.
4	$r \leftarrow r - c(V_{t-1}, D_t)$.
5	If $V_t = \emptyset$, return inconsistent exhaustion.
6	If $ V_t = 1$, return resolved hypothesis.
7	If $r = 0$ and $ V_t > 1$, return exhausted ambiguity $\langle h : h \in V_t \rangle$.

Correctness. At each step, V_t contains exactly the hypotheses that remain compatible with all data seen so far under the loss threshold. The resource budget is decreased by the specified cost. Empty V_t means no compatible hypothesis remains. Singleton V_t means the hypothesis is resolved. If the budget is exhausted while V_t is non-singleton, the system has unresolved ambiguity with no remaining learning resource. $\square$

Algorithm 15.5 (Structural crisis detection). Input: time series $y(t)$, estimated resource $r(t)$, estimated structural Jacobian $D\Omega_t$, and $\varepsilon > 0$.

Step	Operation
1	Estimate $a(t) = \dim \ker D\Omega_t$.
2	Compute availability $h(r(t))$.
3	Estimate depletion speed $-\dot{h}(r(t))$.
4	Compute $\mathcal{C}(t) = a(t) \frac{-\dot{h}(r(t))}{h(r(t)) + \varepsilon}$.
5	Flag crisis if $\mathcal{C}(t)$ exceeds a chosen threshold.

Correctness. The index $\mathcal{C}(t)$ is defined as ambiguity dimension multiplied by normalized resource depletion speed. The algorithm computes exactly this quantity and flags large values according to the chosen threshold. $\square$

16 Structural Tables

Object	Formula	Meaning
Allocation derivative	$D\alpha_\theta$	Hidden allocation sensitivity
Observation derivative	$D\omega_z$	State-resource visibility
Structural Jacobian	$D\Omega = D\omega D\alpha$	Allocation-to-observation sensitivity
Tangent ambiguity	$\ker D\Omega$	Invisible allocation perturbations
Resource velocity	$\dot{r} = F_R$	Instantaneous resource motion
Consumption	$c = -\dot{r}$	Depletion rate
Exhaustion boundary	$h(r) = 0$	Terminal resource frontier
Exhaustion time	$\tau = \inf\{t : h(r(t)) = 0\}$	Remaining continuation
Lifetime gradient	$D_\theta \tau$	Sensitivity of exhaustion time
Silent consumption	$\dot{y} = 0, \dot{r} \neq 0$	Hidden depletion

Table 1: Core operators of the Calculus of Structural Resources.

Field	Allocation Θ	Resource R	Exhaustion boundary
AI/ML	Hypotheses, weights	Data, compute, tokens	Budget exhausted
Statistics	Parameters	Fisher information, sample size	Non-identifiability
Finance	Balance-sheet structure	Liquidity, collateral	Insolvency or run
Economics	Production allocation	Capital, labour, energy	Capacity depletion
Political science	Coalitions, legitimacy	State capacity	Governance failure
Geopolitics	Force posture, resolve	Logistics, alliances	Strategic culmination
Warfare	Units, plans, intent	Ammunition, fuel, manpower	Operational exhaustion
Thermodynamics	Microstates	Free energy	No extractable work
Computation	Configurations	Memory, time, stack	Resource halt

Table 2: Structural resource interpretations across disciplines.

Regime	Mathematical condition	Interpretation
Determinate active	$ F_y = 1, h(r) > 0$	Unique allocation, continuation possible
Ambiguous active	$ F_y > 1, h(r) > 0$	Multiple allocations, continuation possible
Determinate exhausted	$ F_y = 1, h(r) = 0$	Unique allocation, no continuation
Ambiguous exhausted	$ F_y > 1, h(r) = 0$	Multiple allocations, no continuation
Silent consumption	$\dot{y} = 0, \dot{r} \neq 0$	Hidden depletion
Crisis	$a(t) \gg 0, h(r) \approx 0, -\dot{h} \gg 0$	Opaque rapid exhaustion

Table 3: Main regimes of Structural Resource Calculus.

17 PGF/TikZ Illustrations

17.1 Fundamental Diagram

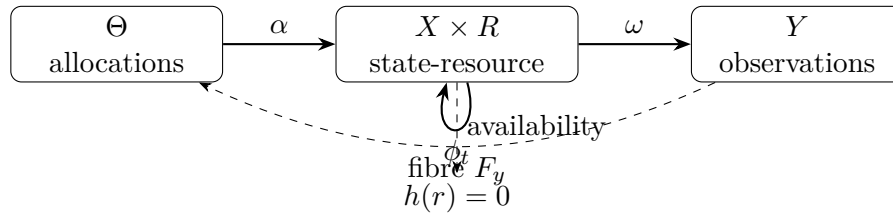


Figure 1: The structural resource diagram. Ambiguity lives in inverse fibres. Exhaustion lives at the resource boundary.

17.2 Tangent Ambiguity

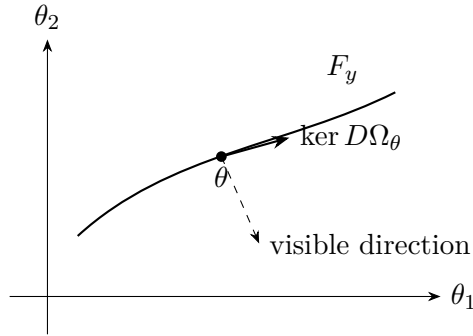


Figure 2: Tangent ambiguity is the kernel of the observation-allocation derivative. Motions along the fibre do not change the observation to first order.

17.3 Resource Descent and Exhaustion

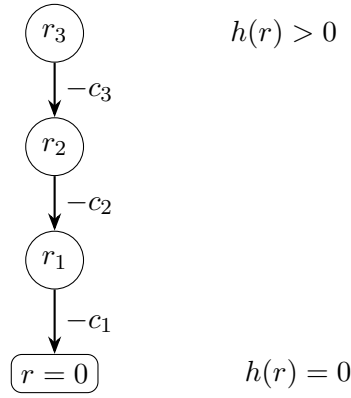


Figure 3: Resource descent terminates at the exhaustion boundary.

17.4 Silent Consumption

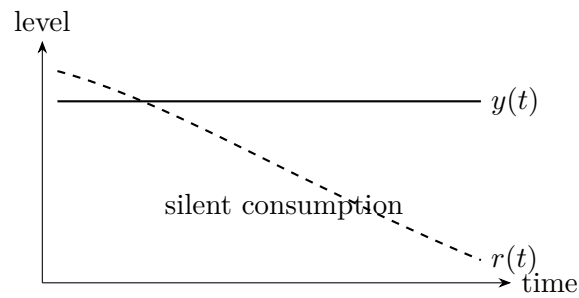


Figure 4: Observation remains constant while hidden resource declines.

17.5 Crisis Geometry

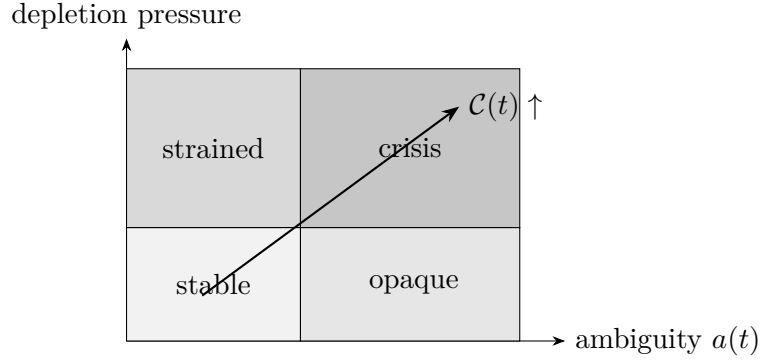


Figure 5: A crisis combines high ambiguity with high normalized resource depletion.

18 Open Problems

- P1. Fibre-boundary duality.** Develop a formal duality between ambiguity fibres F_y and exhaustion boundaries $h(r) = 0$.
- P2. Nonsmooth structural resource calculus.** Build a complete calculus for min, max, threshold, and weakest-link resource systems.
- P3. Stochastic exhaustion.** Extend exhaustion-time sensitivity to stochastic differential systems and Markov decision processes.
- P4. Statistical identifiability.** Relate $\ker D\Omega$ and $\ker I(\theta)$, where $I(\theta)$ is the Fisher information matrix.
- P5. Learning bounds.** Derive sample-complexity and compute-complexity bounds from rates of ambiguity reduction per unit resource.
- P6. Crisis prediction.** Estimate the crisis index

$$\mathcal{C}(t) = a(t) \frac{-\dot{h}(r(t))}{h(r(t)) + \varepsilon}$$

from noisy empirical data.

- P7. Warfare culmination.** Use resource sensitivity formulas to predict operational culmination under uncertain logistics and adversarial action.
- P8. Political exhaustion.** Model collapse of legitimacy, fiscal capacity, or administrative capacity as structural resource exhaustion.
- P9. Thermodynamic semantics.** Relate $h(r)$, free energy, entropy production, and Landauer cost in irreversible computation.
- P10. Category-theoretic calculus.** Construct a categorical differential calculus on **SRes**, with tangent functors, fibre functors, and boundary functors.

19 Conclusion

The Calculus of Structural Resources studies systems that allocate hidden resources, reveal partial observations, evolve dynamically, consume continuation capacity, and terminate at resource boundaries.

Its three central formulas are:

$$D\Omega_t = D\omega_{\phi_t(\alpha(\theta))} D\phi_t|_{\alpha(\theta)} D\alpha_\theta,$$

$$\mathcal{A}_\theta(t) = \ker D\Omega_t(\theta),$$

and

$$D_\theta \tau = - \frac{Dh_{r(\tau)} P_R(\tau)}{Dh_{r(\tau)} F_R(z(\tau), u(\tau))}.$$

The first is the structural chain rule. The second differentiates ambiguity. The third differentiates exhaustion.

Across mathematics, economics, finance, political science, geopolitics, international relations, warfare, statistics, data science, artificial intelligence, thermodynamics, and computation, the same formal pattern appears:

hidden allocation $\rightarrow$ partial observation $\rightarrow$ resource-consuming evolution $\rightarrow$ exhaustion boundary.

Thus:

Structural Resource Calculus is the calculus of finite futures under partial visibility.

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Glossary

Allocation

The hidden initialization of a structural resource system, represented by

$$\alpha : \Theta \rightarrow X \times R.$$

Allocation fibre

The set

$$F_y = (\omega \circ \alpha)^{-1}(y)$$

of hidden allocations compatible with observation y .

Ambiguity

Non-singleton compatibility of hidden allocations with one observation:

$$|F_y| > 1.$$

Ambiguous lifetime profile

The ambi-object

$$\Lambda(A) = \langle \tau_1 \mid \cdots \mid \tau_n \rangle$$

recording branchwise exhaustion times.

Availability function

A smooth map

$$h : R \rightarrow \mathbb{R}$$

whose zero set defines the exhaustion boundary.

Availability predicate

A binary map

$$\chi : R \rightarrow \{0, 1\}$$

declaring whether a resource remains available.

Burn rate

The rate at which a resource is consumed:

$$c(t) = -\dot{r}(t).$$

Crisis index

The quantity

$$\mathcal{C}(t) = a(t) \frac{-\dot{h}(r(t))}{h(r(t)) + \varepsilon},$$

combining ambiguity and normalized depletion pressure.

Culmination

In warfare, the point at which a force no longer has sufficient resources to continue offensive or productive operations.

Dynamic tangent ambiguity

The time-dependent kernel

$$\mathcal{A}_\theta(t) = \ker D\Omega_t(\theta).$$

Exhaustion

The event that resource availability reaches the terminal boundary:

$$h(r) = 0$$

or

$$\chi(r) = 0.$$

Exhaustion boundary

The set

$$\partial R_{\text{ex}} = \{r : h(r) = 0\}.$$

Exhaustion time

The first time at which the resource trajectory reaches the exhaustion boundary:

$$\tau = \inf\{t : h(r(t)) = 0\}.$$

Exhaustion-time sensitivity

The derivative

$$D_\theta \tau = - \frac{Dh_{r(\tau)} P_R(\tau)}{Dh_{r(\tau)} F_R(z(\tau), u(\tau))}$$

measuring how allocation changes exhaustion time.

Fibre entropy

The entropy of the posterior distribution over a finite allocation fibre.

Hidden allocation space

The space Θ of latent causes, priors, hypotheses, strategies, microstates, force postures, or resource allocations.

Learning efficiency

The ambiguity reduction per unit resource:

$$\eta_{\text{learn}}(t) = \frac{-\dot{a}(t)}{c(t)}.$$

Observation-allocation map

The composite map

$$\Omega = \omega \circ \alpha : \Theta \rightarrow Y.$$

Resource

A continuation-enabling quantity, such as memory, time, energy, liquidity, collateral, legitimacy, ammunition, manpower, or compute.

Resource curvature

The acceleration of resource motion:

$$\ddot{r}(t).$$

For consumption $c = -\dot{r}$, accelerating consumption means $\dot{c} > 0$.

Resource monotone

A scalar function that does not increase along admissible resource-consuming transitions.

Resource velocity

The resource component of the structural resource vector field:

$$\dot{r} = F_R(x, r, u).$$

Silent consumption

A regime in which observation does not change but resource is consumed:

$$\dot{y} = 0, \quad \dot{r} \neq 0.$$

State-resource pair

An element

$$z = (x, r) \in X \times R.$$

Structural chain rule

The identity

$$D\Omega_t = D\omega_{\phi_t(\alpha(\theta))} D\phi_t|_{\alpha(\theta)} D\alpha_\theta.$$

Structural Jacobian

The derivative

$$D\Omega_\theta$$

mapping allocation perturbations to observation perturbations.

Structural resource system

The tuple

$$\mathcal{S} = (\Theta, X, R, Y, \Sigma, \alpha, \delta, \omega, \chi).$$

Tangent ambiguity

The kernel

$$\mathcal{A}_\theta = \ker D\Omega_\theta$$

of locally invisible allocation perturbations.

Version-space exhaustion

The termination of learning when the resource budget is depleted before the compatible hypothesis set has been reduced to a singleton.

The End